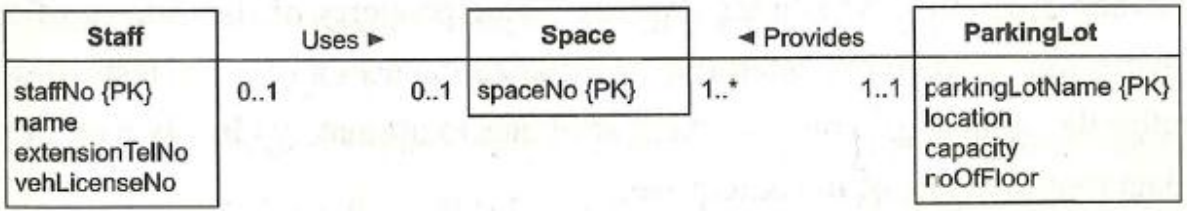
COMP2111 Database Design
Class Practice 3 Entity-Relationship Modeling

1. Create an ER model in UML notation for a large organization having several parking lots used by staff. Represent all the following descriptions in parts (a), (b) and (c) as a single ER model. Provide any assumptions necessary to support your model.
   a) Each parking lot has a unique name, location, capacity, and number of floors (where appropriate).
   b) Each parking lot has parking spaces, which are uniquely identified using a space number.
   c) Members of staff can request the use of a parking space. Each member of staff has a unique number, name, telephone extension number, and vehicle license number.



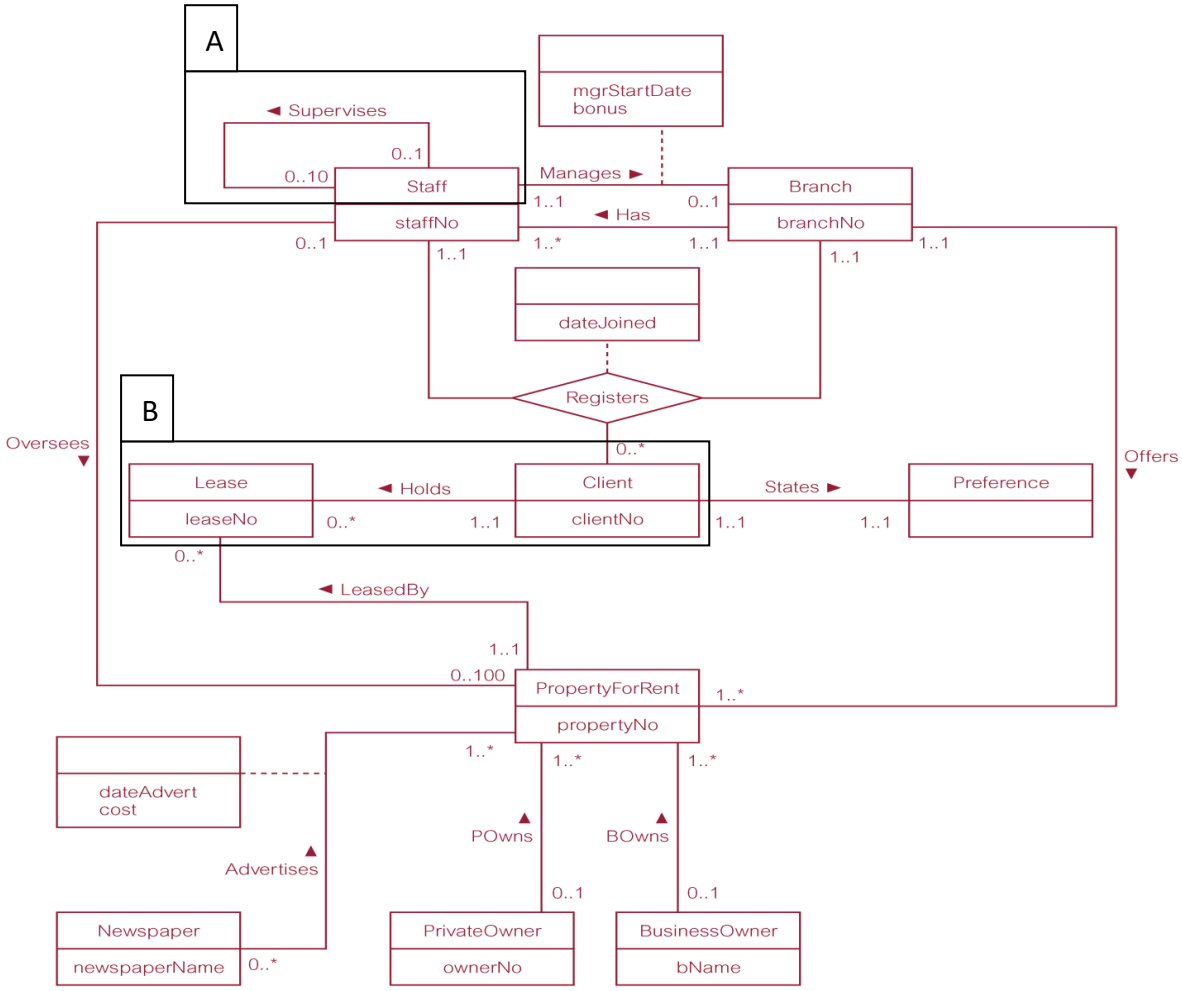
Assumption:
Each parking space can be used by 1 staff only, and some parking space can be left unassigned.

2. Given the following ER diagram of Branch user views of DreamHome,

2.1. Use two sentences to describe the multiplicity for each of the chosen relationships.

2.2. Indicate the structural constraint for each of the chosen relationships. Which one is it?

- one-to-one (1:1)
- one-to-many (1:*)
- many-to-many (*:*)



(A): Multiplicity

- Each staff can supervise **zero to 10** members of staff.
- Each staff is supervised by **zero or 1** member of staff.

Structural constraint: This is a **1:*** relationship.

(B) Multiplicity

- Each client holds **zero or more** leases.
- Each lease is held by **one and only one** client.

Structural constraint: This is a **1:* relationship**